

A High-Performance Miniaturized Frequency Shift Detection System for QCM-Based Gravimetric Sensing

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Micro-gravimetric sensors, such as quartz crystal microbalance (QCM) are capable of detecting trace substances and even single nanoparticles in various fields due to their high sensitivity, selectivity, and stability. As a resonant sensor, the detection response of QCM requires frequency analysis instruments with high precision like a frequency counter and a vector network analyzer to measure its frequency shift during mass sensing. However, such bulky and high-cost instruments undoubtedly hinder the applications of QCM-based sensors outside the laboratory such as in situ and portable detection. In this paper, a high-performance miniaturized frequency shift detection system based on a phase-locked loop (PLL) circuit, is developed for QCM measurement. The designed system achieves high detection sensitivity of frequency shift as 1.859 mV Hz^{-1} with the Allan deviation of 0.49 mV at 0.15 s and frequency resolution of 0.26 Hz , which is conducive to achieving the detection of trace substance by QCM. The excellent measuring performance is further validated by measuring the frequency response of QCM during mass sensing in a gaseous environment and aqueous solution. As a result, compared with a commercial frequency counter, the superior linearity and accuracy of over 98.4% were confirmed with a mean relative error (MRE) of less than 0.92%.

the gaseous environment and aqueous solution as electronic noses for mass spectrometry in a wide variety of fields, including industrial gas sensing,^[3–5] disease diagnosis,^[6,7] and environmental pollutant monitoring.^[8,9] Based on the frequency response, the micro-gravimetric sensors can realize specific detection of various substances by surface functionalization, which is equipped with excellent sensitivity, specificity, and environmental stability, attracting a great deal of research interest at present.^[10–12] The frequency shift response of QCM toward micro-mass detection provides high accuracy and stability.^[13] However, due to its high nominal resonant frequency and relatively narrow frequency shift range during mass sensing, the high-performance frequency analysis capability of analyzers such as vector network analyzers, frequency counters, and frequency spectrum analyzers with high precision are essential instruments for QCM frequency response

measurement. Compared to millimeter-sized sensor devices, the bulky and expensive analyzer undoubtedly hinders their development of miniaturization, commercialization, and multi-scene applications like in situ detection.^[14,15]

According to the characteristics of QCM, two important features for its frequency response analysis need to be considered in mass detection. The first one is the high nominal resonant frequency of QCM at a level of 10^6 – 10^8 Hz , the higher frequency

1. Introduction

Micro-gravimetric sensors based on micro/nano-electro-mechanical systems (M/NEMS), such as quartz crystal microbalance (QCM), film bulk acoustic resonator (FBAR), and nano-resonators are capable of detecting trace substances, even down to single nanoparticles,^[1,2] and have been applied in

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DOI: 10.1002/adsr.202400148

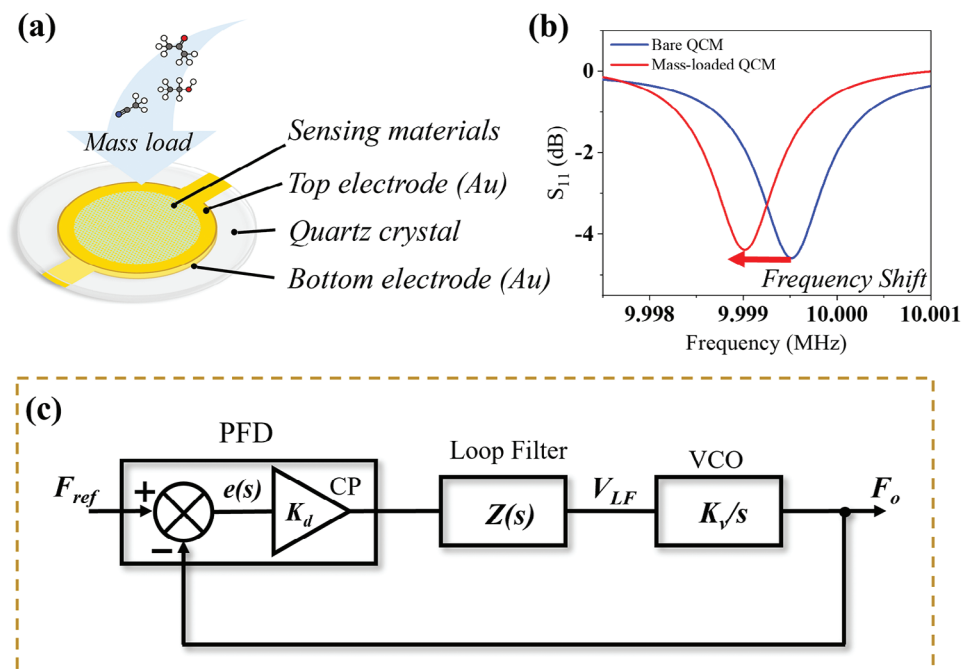


Figure 1. The detection principle of QCM: a) the structure of QCM and mass loading schematic diagram. b) Working principle of QCM mass sensor. c) The diagram of a typical PLL model.

corresponds to the higher mass sensitivity.^[16,17] On the other hand, during the process of mass detection, the frequency shift response of QCM can be as minimal as 1 Hz, and the common range of frequency shift response is 1–2000 Hz, which corresponds to ppm-level of its nominal resonant frequency.^[18–21] Due to the characteristics mentioned above, the frequency response analyzers for QCM need to be equipped with high-frequency measurement capability and high precision to distinguish variations of a few Hz, which is the challenge lies in miniaturizing and simplifying frequency response analyzer for QCM. The aforementioned instruments achieve high-performance frequency analysis through complex internal components with high quality. The power supply, cooling system, and human-machine interfaces within these instruments contribute to their bulky size and high cost. Besides, the present miniaturized frequency analyzers have sacrificed some performance, resulting in a lack of ppm-level frequency resolution capabilities in the high-frequency operating range.^[22,23]

In the current study, a phase-locked loop (PLL) provides a frequency-shift detection solution with high precision and stability by locking frequency and phase. The PLL-based system can reduce noise by adjusting circuit bandwidth while maintaining frequency tracking, enhancing the accuracy of frequency detection. The flexibility of PLL allows the system's speed and precision to be balanced according to application requirements, contributing a significant advantage in tracking QCM frequency changes in real-time.^[22,24] In this work, a miniaturized frequency shift detection system for QCM measurement based on a PLL has been proposed for sensing applications. The detection principle of the QCM-based mass sensor and the working principle of the PLL circuit-based frequency shift detection system are introduced. According to the detection requirements of the QCM sen-

sor, the proposed miniature system can operate at the same high-frequency level as the QCM resonant frequency and is equipped with performances including low noise, high sensitivity for frequency shift-DC voltage converting, linearity, and accuracy, contributed by designing the crystal oscillator and signal processing module with high-quality components. Further, the proposed system was validated through detection experiments of volatile organic compounds (VOCs) and heavy metal ions (HMIs) using QCM, the frequency response of which was measured by both a commercial frequency counter and the developed system. By comparing the measurement results, the superior linearity and accuracy were confirmed. Combined with its small size and low cost, the developed system has the potential to become the optimal measurement instrument for QCM in situ and portable applications.

2. Results and Discussion

2.1. The Principle of Detection

The principle of QCM for micro-mass detection is illustrated in Figure 1a,b. Based on the piezoelectric effect, when QCM is driven by an external excitation signal, a resonance oscillation occurs at a fixed frequency, which is related to the crystal thickness.^[13] Meanwhile, the vibration mode of QCM is sensitive to any surface perturbations, which can be described as a negative resonant frequency shift due to the micro-mass loaded or adhered to the surface of QCM, as illustrated in Figure 1b and explained by the Sauerbrey equation:^[25]

$$\Delta f = -\frac{2f_0^2}{A\sqrt{\rho_q u_q}} \Delta m \quad (1)$$

where f_0 and Δf represent the resonant frequency of QCM and its frequency shift caused by mass loading Δm , respectively; μ_q and ρ_q represent the elastic modulus and density of the piezoelectric quartz crystal, respectively; and A is the effective area. Equation (1) shows a linear relationship between frequency shift and the mass loaded to the surface of QCM. Therefore, the key parameter of a QCM-based micro-mass sensor in mass detection is the change in its resonant frequency.

PLL is a circuit that maintains a generated signal in a fixed phase relationship to a reference signal, the output of the PLL can trace the frequency of the input signal with high resolution. A typical PLL model is shown in Figure 1c, which consists of a phase-frequency detector (PFD) with a charge pump, a loop filter, and a voltage-controlled oscillator (VCO), which can be replaced by a voltage-controlled crystal oscillator (VCXO) with high-frequency accuracy, stability, and low phase noise.

An oscillation signal generated by QCM as reference frequency signal (F_{ref}) inputs into the PLL system, and is compared with the output frequency signal (F_o) by PFD, which generates an error signal ($e(s)$) according to their frequency or phase difference. Based on the error signal, the loop filter outputs a smooth voltage signal (V_{LF}) that controls the frequency of the oscillation signal (F_o) generated by VCO to keep it consistent with the frequency of the oscillation generated by QCM. When F_{ref} and F_o maintain a fixed phase relationship, the loop is locked and reaches a stable state. The transfer function of the PLL can be described by:

$$\frac{F_o}{F_{ref}} = \frac{K_d K_v Z(s)}{s + K_d K_v Z(s)} \quad (2)$$

where K_d and K_v are PFD gain and voltage-controlled gain, respectively. In the locked state of the loop, the loop filter's signal (V_{LF}) is linearly related to the oscillation frequency, so that V_{LF} can be used as the system output to track the frequency shift of the QCM. Accordingly, the transfer function can be obtained as follows:

$$\frac{V_{LF}}{F_{ref}} = \frac{s K_d Z(s)}{s + K_d K_v Z(s)} \quad (3)$$

which means that V_{LF} can be described quantitatively as a function of F_{ref} .

2.2. The Circuit Design

In order to design a miniaturized device for frequency shift detection, the proposed circuit consists of a crystal oscillator, a PLL, a low-pass filter, and an inverting amplifier. The design of each part is described in detail as follows.

A Pierce oscillator is applied to drive QCM to generate an oscillation signal as reference frequency (F_{ref}), and the oscillator circuit is shown in Figure 2a. The Pierce oscillator, satisfying the Barkhausen oscillation condition, forms a positive feedback loop to generate an oscillation signal with a stable frequency, which is the resonant frequency fixed by the crystal due to its frequency selection property. The end resistor (R_3) of the circuit is used to perform high-frequency filtering on the output square wave oscillation signal, reducing noise and stabilizing the waveform.

The phase-frequency detector with a charge pump is obtained by a high-performance PLL chip (TLC2933, Texas Instruments), the schematic diagram of which is shown in Figure 2b. Two D flip-flops are employed to compare the phase and frequency disparities of the two input signals (F_{ref} & F_o). Their Q outputs are utilized to control the charging and discharging of the charge pump and to generate the error signal. As shown in Figure 2c, the loop filter is designed as a second-order lag-lead filter with a narrow bandwidth (<100 Hz) to integrate the error signal, suppressing high-frequency noise. The transfer function of the loop filter can be described as:

$$Z(s) = \frac{1 + s \cdot R_4 C_1}{1 + s \cdot (R_4 C_1 + R_3 C_1)} \quad (4)$$

The cut-off frequency of the loop filter is set at ≈ 22.5 Hz, as illustrated by the bode plot in Figure 2d, which shows a straight line with a slope of -20 dB dec^{-1} after the cut-off frequency point, filtering out the interference of high-frequency noise and leading great stability to the PLL circuit system. According to the frequency shift range of QCM commonly in mass detection, a MEMS-based VCXO (SiT3808, SiTime) with a pull range of 400 ppm is applied. The selected high-performance VCXO works at a high basic resonant frequency of 10 MHz, which depends on the resonant frequency of the QCM. The voltage-controlled gain (K_v) of the VCXO was 2.766 Hz mV^{-1} , with high-frequency stability, and low phase noise of -160 dBc Hz^{-1} at high-frequency background near 10 MHz. When the loop is locked, the frequency of the signal outputted by the VCXO is consistent with current oscillation frequency of the QCM, as shown in Figure S1 (Supporting Information), which can be represented by the control signal (V_{LF}) of the loop filter to the VCXO.

After that, a second-order Sallen-key filter with a voltage follower is employed to suppress the noise, protect the circuit, and stabilize the signal, as shown in Figure 2e, where the voltage follower is used to isolate the signal from external interference. In order to make the system's output signal (V_o) positively correlated with mass loaded on the QCM and enhance its sensitivity in frequency shift detecting, an inverting amplifier is introduced due to the negative relationship between QCM's frequency shift and mass loading. A reference voltage (V_{ref}) controlled by the micro-controller is set to subtract with the filter output (V_{LP}) for signal inverting. The V_{ref} needs to be adjusted to a certain value to eliminate small differences in the fundamental frequency between different QCMs. The amplification gain (A_{amp}) of the amplifier depends on the amplification resistor (R_{amp}), which is designed as 25 k Ω , resulting in A_{amp} as 5. The output of the inverting amplifier can be obtained by:

$$V_o = A_{amp} \times (V_{ref} - V_{LP}) \quad (5)$$

As shown in Figure 2f, the simulation of the filter and inverting amplifier illustrate the filter's signal noise reduction and stabilization effects, and improve the frequency shift detection sensitivity by amplifying the output signal. Figure 2g shows the main working principle of the designed frequency shift detection system. It illustrates that under the loop-locked condition, the frequency of the QCM and VCXO's output signal is consistent, and a decrease in their frequency is reflected by an increase in the

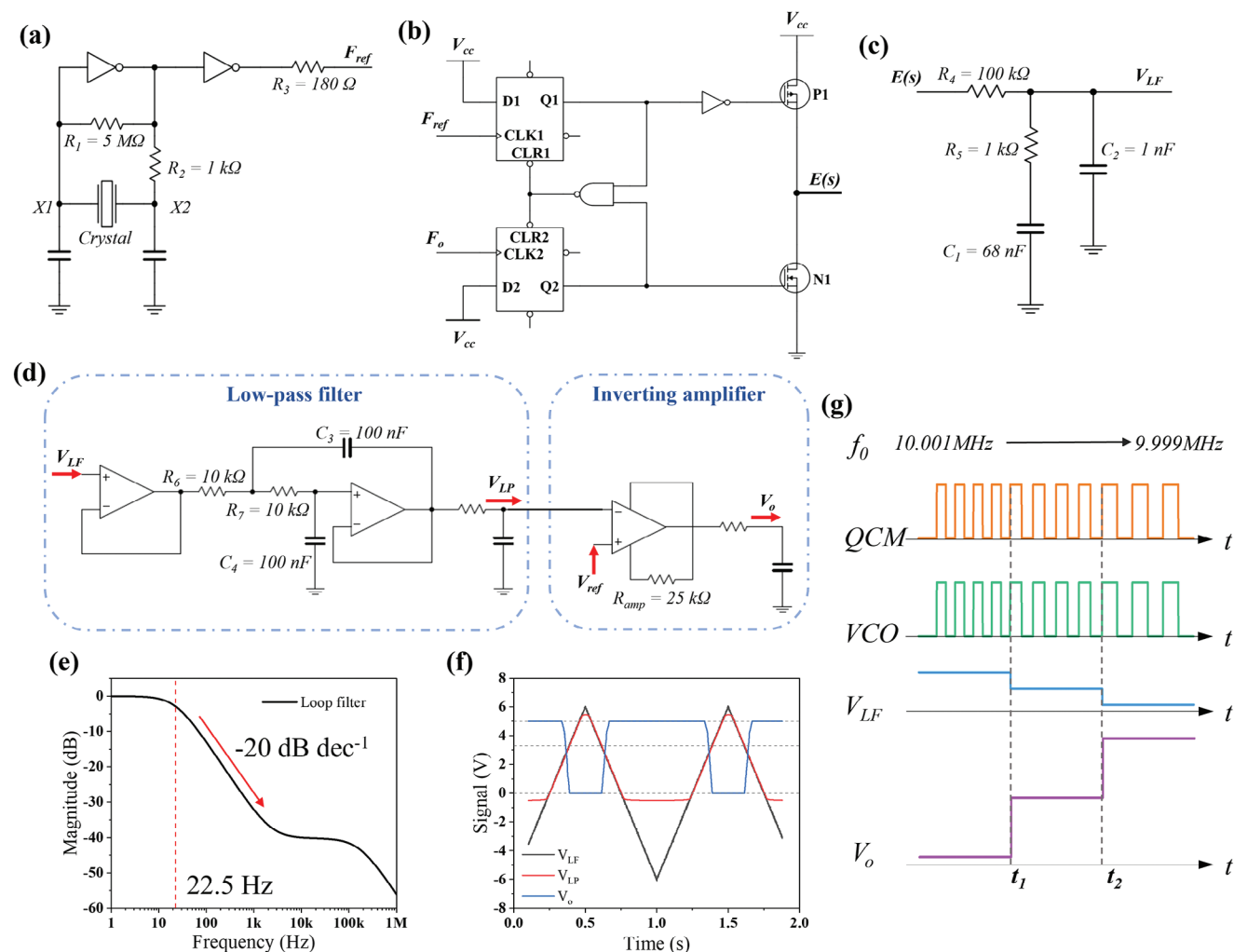


Figure 2. The configuration of a) pierce oscillator, b) phase-frequency detector, c) loop filter. d) The bode plot of the designed loop filter. e) The configuration of the low-pass filter and inverting amplifier. f) The simulated signal of the low-pass filter and inverting amplifier. g) Main working principles of the system.

output signal, so as to realize the detection of the frequency shift. Accordingly, throughout this system, the high-performance PFD and MEMS-based VCXO with ultra-low noise and high Q-factor provide key device stability. The optimally designed low-noise LPF and amplifier offer high sensitivity for frequency shift detection, which ensures that the proposed system has high precision in measuring the frequency response of QCMs.

2.3. System Realization and Testing

The block diagram of the developed frequency shift detection system is shown in **Figure 3a**, and the components and their models are listed in **Table S1** (Supporting Information). **Figure 3b** shows the image of the frequency shift detector, designed as a print circuit board (PCB) with a size of 40 mm×40 mm. The loop filter voltage (V_{LF}), as the key signal in the system, is related to the current oscillation frequency of QCM and VCXO with the loop-locked state, as expressed in: $F_{QCM} = F_{VCXO} = K_v \cdot V_{LF}$. Thereby, the change of loop filter voltage is linearly related to the frequency

shift of QCM, which can be described as: $\Delta F_{QCM} = \Delta V_{LF} \cdot K_v^{-1}$. Thereby, the output of the system (V_o) can be described as:

$$\Delta V_o = A_{amp} \cdot K_v^{-1} \cdot \Delta F_{QCM} \quad (6)$$

The system sensitivity ($\Delta V_o / \Delta F_{QCM}$) can be enhanced by increasing the amplification gain (A_{amp}) and decreasing the VCXO gain (K_v). While there is a constraint in this system that prevents the variation of A_{amp} and K_v . Since the maximum output voltage is limited by the supply voltage of 5 V, the linear tunable range of frequency shift becomes smaller with the sensitivity increases. Thus, an optimum tunable range should be estimated according to the range of frequency shift during QCM mass detection, so that the highest possible sensitivity can be adjusted.

In the test, with the loop locked, the oscillation signal frequency of the QCM and VCXO were kept consistent, as shown in **Figure 3c**. In this work, a VCXO with a pull range of ± 4 kHz and voltage-controlled gain of 2.766 Hz mV^{-1} is selected, which is equipped with a wide enough tunable range for QCM-based

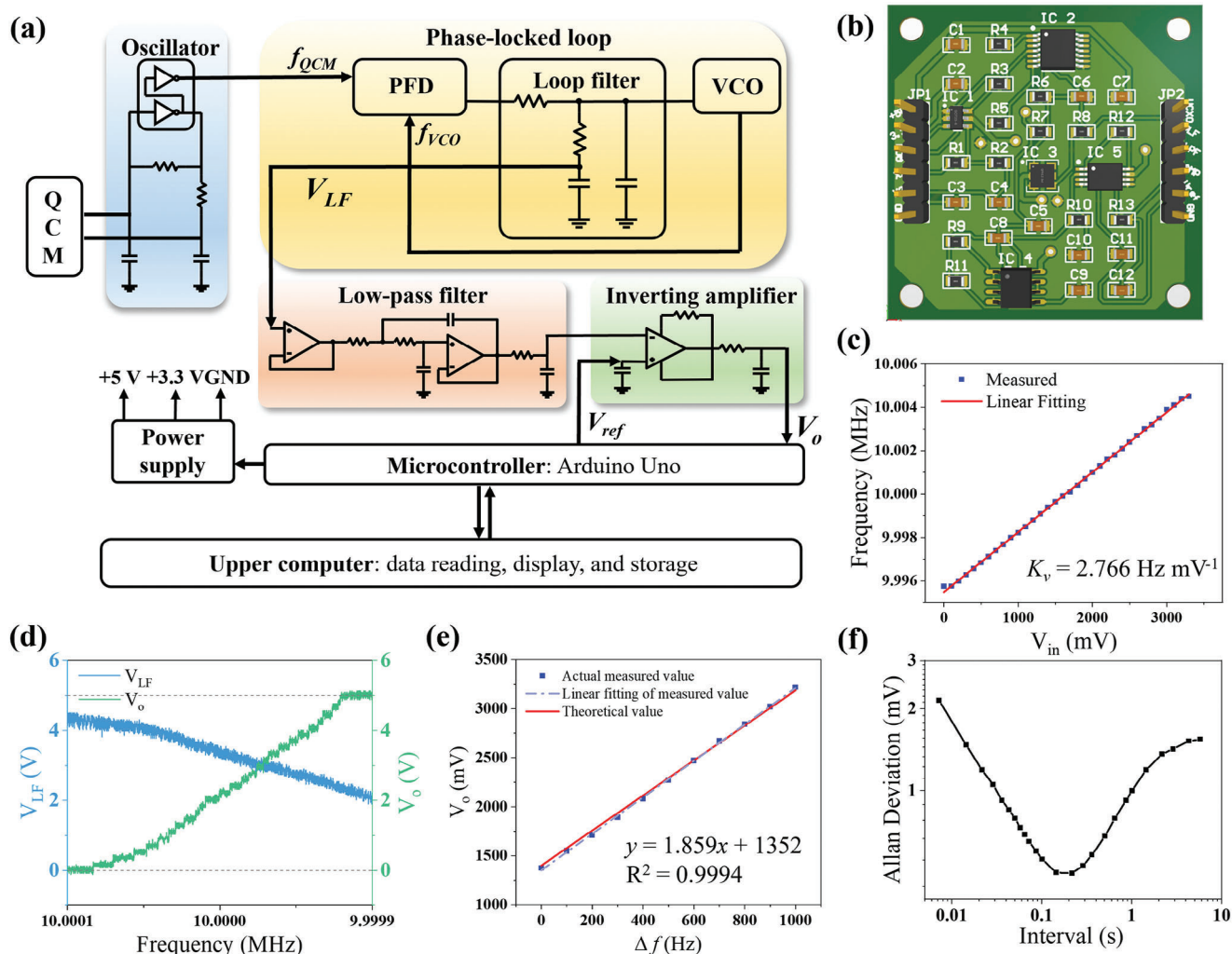


Figure 3. a) The block diagram of the developed frequency shift detection system. b) The PCB figure of the developed detector. c) The frequency-voltage characteristic curve of the VCXO. d) The measured signal of the loop filter (V_{LF}) and the amplifier (V_o) with frequency decline. e) The theoretical measured voltage-frequency characteristic curve. f) Allan deviation plot.

mass detection. Meanwhile, the amplification gain is set as 5. Thus, the sensitivity of the developed frequency shift detector can be calculated as 1.808 mV Hz^{-1} by Equation (6) theoretically, with the measurable frequency shift range of 0–2000 Hz. A frequency generator was employed to take the place of QCM to produce an oscillating square wave near 10 MHz with a decreasing step of 100 Hz for calibrating the sensitivity of the proposed detection system. As shown in Figure 3d, as the frequency of the input signal decreases, the loop filter output (V_{LF}) voltage decreases, while the system output (V_o) increases, which is an amplification of the former. The actual frequency shift detection sensitivity of the system was obtained by linear regression of the measured data, as presented in Figure 3e. The experimental sensitivity was calculated as 1.859 mV Hz^{-1} , which is slightly higher than the theoretical result, which may be attributed to the experimental environment temperature and the error introduced by the instruments during the measurement.

In order to analyze the noise levels and frequency stability during the frequency shift detection, the Allan deviation was intro-

duced, which as a noise analysis method was used for frequency-based sensors and devices widely, as shown in Equation (7):^[26,27]

$$\sigma(\tau)^2 = \frac{1}{2(k-1)} \sum_{i=1}^{k-1} (\bar{y}_{i+1} - \bar{y}_i)^2 \quad (7)$$

where σ represents the Allan deviation, τ is a time interval of measurement, and $\bar{y}_i$ is the mean value of the measured signals during the k -th time interval. In this work, the data sampling rate is related to the program complexity of the microcontroller. When only the system output was collected, the sampling rate of the microcontroller was $\approx 138 \text{ Hz}$. Figure 3f shows a standard Allan deviation plot with a “U-shaped” curve. The Allan deviation initially decreases with increasing interval time of measurement, indicating the noise level tends to stabilize as interval time extends. With a time interval between 0.15 and 0.2 s, the Allan deviation plot shows a flat noise floor, showing a minimal noise deviation of 0.49 mV. As a frequency shift-voltage conversion system, the frequency resolution can be obtained by dividing the Allan de-

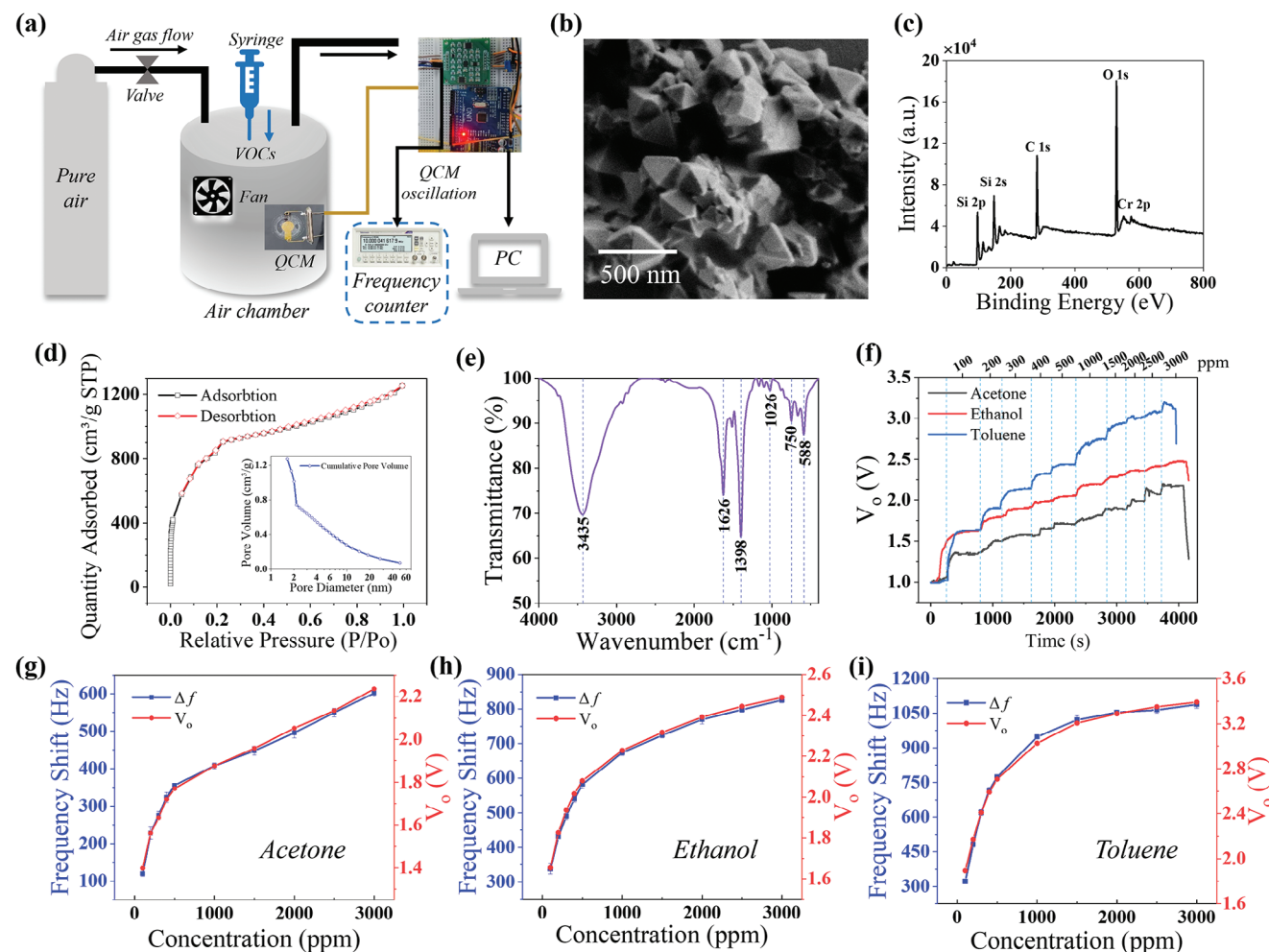


Figure 4. a) The setup of VOC testing with the developed detector. b) The SEM image of MIL-101(Cr). c) The XPS spectra of MIL-101(Cr). d) N₂ adsorption/desorption isotherm and BJH cumulative pore volume of MIL-101(Cr). e) FT-IR transmittance spectra of MIL-101(Cr). f) Dynamic sensing performance in a concentration range of 100–3000 ppm for acetone, ethanol, and toluene. The response comparison between the frequency counter and developed detector for g) acetone, h) ethanol, and i) toluene sensing by QCM.

viation by the sensitivity of the system,^[22] resulting in a resolution of 0.26 Hz. As an important parameter of frequency stability analysis for the frequency system, the Allan deviation of the proposed system is comparable with other similar systems in recent literature,^[22,28–31] as shown in Table S2 (Supporting Information). The features of the developed system along with comparison with other works are summarized in Table S3 (Supporting Information).

2.4. VOC Testing

In order to investigate the detection capability of the developed frequency shift detector in QCM mass detection, three types of VOCs (Acetone, Ethanol, and Toluene) commonly found in industrial environments are selected for testing with a concentration range from 100 to 3000 ppm. The gas sensing system was built as shown in Figure 4a, where the QCM under test was connected to the proposed detection system. To verify the detec-

tion accuracy of the QCM frequency response, a commercial frequency counter (FCA3100, Tektronix) was used to simultaneously detect the oscillation frequency of the QCM as a reference for the true value. The QCM was placed into a 6 L stainless steel air chamber to shield it from external environmental interference. VOC vapors tested in this work were formed by volatilizing injected VOC solution with a syringe into the chamber. The concentration of volatilized VOC vapor in the chamber can be determined as:^[32]

$$c = \frac{22.4 \times 10^3 \cdot T \rho v}{273 \cdot M_w V} \quad (8)$$

where T is the environment temperature, ρ is the density of VOC in the liquid phase, v is the volume of the injected VOC solution in μL , M_w is the molar mass of VOC in g mol^{-1} , and V represents the volume of the experimental air chamber. A fan placed inside the chamber was used to promote the volatilization of VOC and uniform distribution of vapors. The chamber

and the QCM were purged by dry pure air to ensure that the QCM sensor was in a stable initial state before testing. A type of metal-organic framework (MOF) named MIL-101(Cr) is selected as a sensing material due to its superior gas adsorption capacity, which is widely applied in VOC detection.^[33–35] The surface morphology of MIL-101(Cr) is shown in Figure 4b, which appears as the shape of a regular octahedral crystal with a diameter of 250–400 nm. Figure 4c shows the elemental analysis of MIL-101(Cr). The peak at 575 eV represents the Cr element, which provides the crystal structure support with high stability as the core metal node, bringing superior mechanical strength and porous structures inside the crystal. The N₂ adsorption/desorption isotherm of MIL-101(Cr) as shown in Figure 4d, illustrates its superior porosity and adsorption/desorption capacity with a BET-specific surface area of 3067.8 m² g^{−1}, which is beneficial to adsorb VOC vapors. Figure 4e shows the FT-IR spectra of MIL-101(Cr), where the band at 588 cm^{−1} represents the connected bond between the core metal node (Cr) and organic components, which is the essential condition to form a metal-organic framework. The presence of the dicarboxylate linker in MIL-101(Cr) can be found by the bands at 1626 and 1398 cm^{−1}, which correspond to the framework vibrations of O—C—O and C=O bonds.^[34,36,37]

The dynamic sensing response curves of the proposed detector for acetone, ethanol, and toluene are shown in Figure 4f, illustrating a continuous rise in the response signal with increasing concentrations of VOC. The comparison curves of QCM response to VOCs, measured by a frequency counter and the developed detector, are shown in Figure 4g–i. Each data point was obtained by the average value of measured data tested for five replicate experiments with error bars representing the corresponding standard deviation, the maximum of which is 11.7 mV during the multiple measurements, showing superior repeatability. As a result, the response of the proposed detector to VOC is very close to that of the frequency counter, with a mean relative error (MRE) of 0.66% and an accuracy of 98.5%. Figure S2 (Supporting Information) shows the relationship between the system output and the frequency shift measured by the frequency counter in VOC detection, which exhibits superior linearity with R² exceeding 0.995 and the frequency shift-voltage conversion sensitivity of 1.805 mV Hz^{−1}, matching with the theoretical result well. By detecting different kinds of VOCs with low errors, the high accuracy of the sensor in gas detection can be ensured. The sensitivity of QCM for VOCs with a concentration range of 100–500 ppm measured by frequency shift detector is calculated as 0.90, 1.04, and 2.05 mV ppm^{−1} for acetone, ethanol, and toluene, respectively. The high sensitivity of QCM to VOCs is attributed to the adsorption capacity of MIL-101(Cr) as well as the frequency shift detection sensitivity of the developed detector. However, in the high concentration range, the increase of frequency response of the QCM to VOC slows down as an increasing concentration of VOCs, which is caused by approaching saturation adsorption of MOF toward VOCs.

2.5. Heavy Metal Ion Testing

In the aqueous solution, the proposed detector was used to detect HMI, such as Hg²⁺ and Cd²⁺, commonly regarded as serious pollutants in the ecological environment. In the same manner, the

frequency changes of the QCM were measured simultaneously using the proposed detection system and the frequency counter for comparison. The experimental system is built as shown in Figure 5a, where the QCM was installed in a liquid chamber made of polymethyl methacrylate (PMMA) with 60 μL internal volume. A syringe with a certain concentration of HMI solution was mounted on the syringe pump, connected to the liquid chamber by a hose, and used to introduce a steady liquid flow with 30 μL min^{−1}. A peristaltic pump was used to flush the QCM and liquid chamber between tests with phosphate-buffered saline (PBS) by 200 μL min^{−1}.

Likewise, a type of MOF, named MOF-808, was selected as a sensing material to coat the surface of QCM for HMI testing, due to its strong adsorption performance for HMI.^[38,39] Further, to enhance the efficiency of adsorption capacity for Hg²⁺, MOF-808-SH was synthesized by grafting thioglycolic acid on MOF-808.^[40] Their SEM images are shown in Figure 5b,c, which presents the shape of a regular octahedral crystal with a diameter of 200–300 nm, demonstrating that the synthesized MOF-808-SH retains its original structure without deformation. Figure 5d shows the FT-IR spectra, with a band at 2503 cm^{−1} in MOF-808-SH corresponding to the stretching vibration of sulfhydryl (−SH), confirming its substitution on the structure of MOF-808 successfully.^[41] In Figure 5e, the XPS spectra of MOF-808-SH also show a characteristic peak of 164 eV, corresponding to the S 2p of thioglycolic acid, proving that the sulfhydryl groups were formed successfully in the synthesized MOF-808-SH.^[42]

Hg²⁺ and Cd²⁺ with concentrations of 5, 25, 100, 250, and 500 ppm were measured by MOF-808-SH coated QCM under the experimental conditions, and the responses were collected by the proposed frequency shift detector and frequency counter for comparison. Figure 5f,g shows that the response data of QCM to HMI collected by the developed detector is very close to that collected by the frequency counter, with an MRE of 0.92% and an accuracy of 98.4%. The great linear relationship between the system output and the frequency shift with R² over 0.998 is shown in Figure S3 (Supporting Information), demonstrating that the proposed system possesses the same excellent performance compared with the frequency counter and can be well applied for QCM measurement in a liquid environment. As shown in Figure 5f,g, due to the affinity of sulfhydryl groups in MOF-808-SH to Hg²⁺, the sensitivity of QCM for Hg²⁺ with a concentration range of 5–100 ppm measured by the proposed system is calculated as 9.79 mV ppm^{−1}, which is higher than that for Cd²⁺ (2.03 mV ppm^{−1}). In addition, the selectivity of the QCM functionalized by MOF-808-SH for different cations was assessed and measured by the developed system. Individual ions such as Cu²⁺, Pb²⁺, Cd²⁺, Hg²⁺, and multi-cation mixture were tested in a concentration of 100 ppm, where the mixture contains several common background cations like Na⁺, K⁺, Ca²⁺, Mg²⁺, and NH₄⁺. As shown in Figure 5h, the MOF-808-SH coated QCM exhibits significant detection capacity toward Hg²⁺, compared with the multi-cation mixture, Cu²⁺, Pb²⁺, and Cd²⁺. Meanwhile, the detection capacity toward Hg²⁺ will not be interfered with by other cations in the solution background, with a detection error of less than 2.5%. The results show that the response signal measured by the proposed detection system can quantitatively reflect the concentration of HMI with high accuracy, which has great potential to be applied to various QCM detection scenarios.

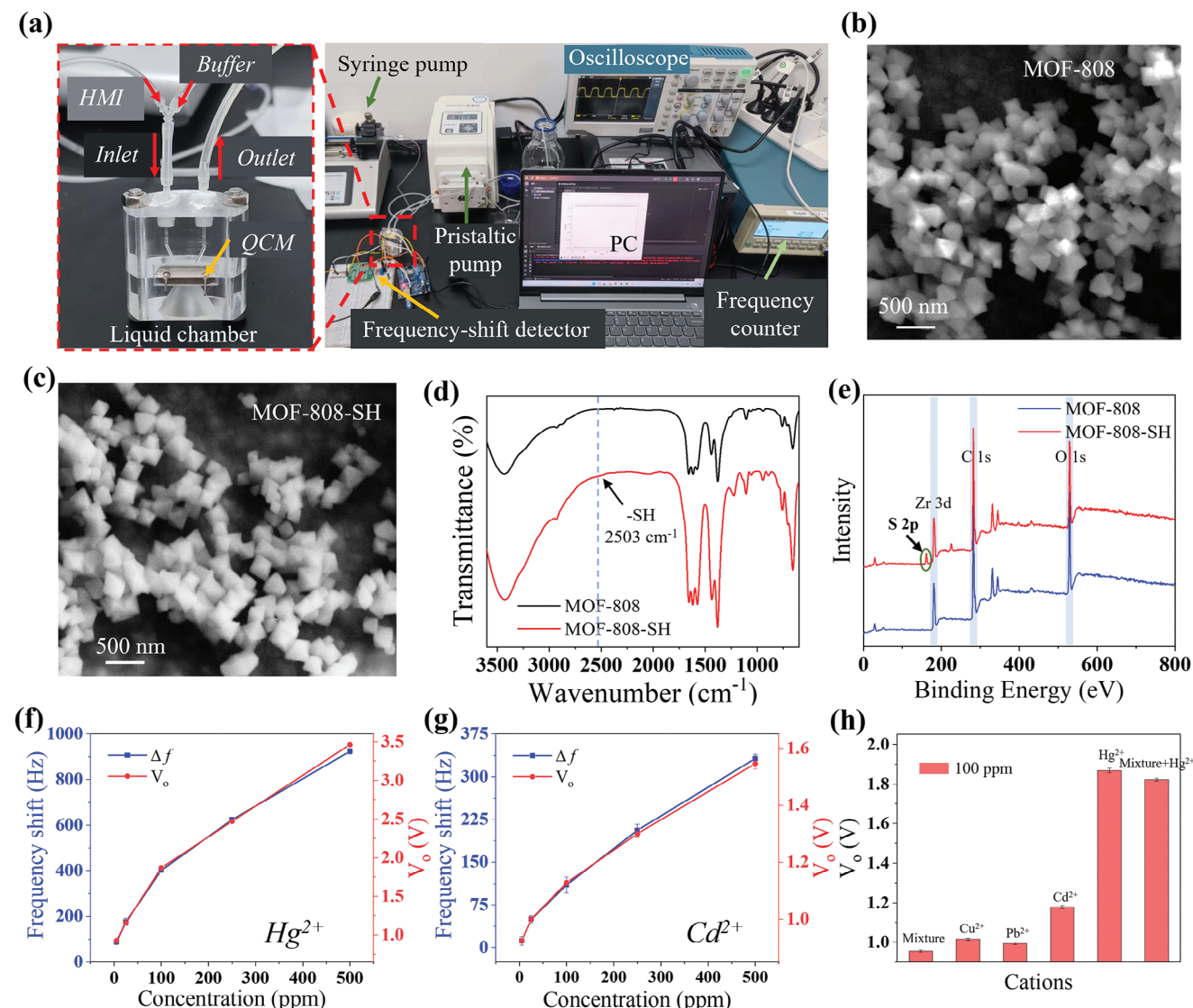


Figure 5. a) The setup for HMI sensing and liquid chamber for testing. The SEM image of b) MOF-808 and c) MOF-808-SH. d) The FT-IR spectra of MOF-808 and MOF-808-SH. e) The XPS spectra of MOF-808 and MOF-808-SH. The response comparison between the frequency counter and developed system for f) Hg^{2+} and g) Cd^{2+} . h) The selectivity of sensing material coated QCM system for different cations.

3. Conclusion

In this work, a miniaturized frequency shift detection system for QCM measurement was developed with a small size and low cost. Based on a PLL circuit, combined with a Pierce oscillator as a QCM driver, a low-pass filter, and an inverting amplifier, the developed system was equipped with a high-frequency shift-DC voltage converting sensitivity of 1.859 mV Hz^{-1} . The frequency shift detection range of the proposed system is 0–2000 Hz, which can meet the detection requirements of QCM in mass sensing. Through the detection experiments of VOC and HMI by QCM, the ability of the proposed system in frequency shift detection is comparable to that of commercial frequency counter, showing strong frequency shift analysis performances with low MRE of 0.66%, high accuracy of 98.4%, and high linearity. The tests for the proposed system in a gas environment and aqueous solu-

tion demonstrate its potential employment in mass sensing with QCM in the application of in situ and portable detection. In the future, this technology can also be applied to other frequency shift-based sensing devices such as FBARs. The proposed system can also be further integrated into small electronic devices to simplify the measuring methods for QCM and improve its large-scale application.

4. Experimental Section

Materials: MIL-101(Cr) and MOF-808(Zr) were purchased from XF-NANO. Thioglycolic acid, and *N,N*-dimethylformamide (DMF) were purchased from Sigma-Aldrich. Acetone, ethanol, toluene (HPLC grade), ammonia liquor ($\text{NH}_3 \cdot \text{H}_2\text{O}$, 28%), and hydrogen peroxide (H_2O_2 , 30%) were purchased from Sinopharm Chemical Reagent. Mixed cations standard so-

lution (Na^+ , K^+ , Ca^{2+} , Mg^{2+} , and NH_4^+ , 500 mg L^{-1} in water), Mercury, cadmium, cuprum, and plumbum ion standard solution (Hg^{2+} , Cd^{2+} , Cu^{2+} , and Pb^{2+} , 500 mg L^{-1} in water) were purchased from Macklin. PBS was purchased from Biosharp. The 10 MHz AT-cut QCMs with Cr/Au electrodes were purchased from Yantai Kesheng Electronics.

Synthesis of MOF-808-SH: The synthesis process of MOF-808-SH was prepared as illustrated in Figure S4 (Supporting Information).^[40] The MOF-808 was activated at 150 °C overnight before further modification. 2 mL of thioglycolic acid was dissolved in 10 mL of DMF solution, then 0.1 g of activated MOF-808 was added, followed by sonication for 30 min to ensure dispersion well. The mixture was heated to 70 °C for 24 h under a nitrogen atmosphere. Then, the modified MOF-808 was washed with DMF and acetone solution several times, collected by centrifugation, and dried overnight in a vacuum over 60 °C.

Sensor Functionalization: MIL-101(Cr) and MOF-808-SH were selected as sensing materials for VOC and HMI detection, respectively. AT-cut QCMs with the basic resonant frequency of 10 MHz were washed with ammonia liquor, hydrogen peroxide, and DI-water (1:1:5, v/v/v) for 10 min in 70 °C, followed by flushed with DI-water and dried. Then, the cleaned QCMs were treated in low-pressure oxygen plasma for 10 min to remove the organic contaminants and create a hydrophilic interface on its backside, which facilitates the even spread of MOF suspension. Meanwhile, MIL-101(Cr) and MOF-808-SH nanoparticles were mixed with pure ethanol solution at 0.2% w/v followed by 30 min blending to form a suspension.^[43] After that, a 5 μL suspension was drop-coated onto the surface of QCM several times, forming a sensitive layer. There was an ≈ 5 min interval to allow the evaporation of the excessive solvent. The MOF-coated QCMs were heated to ≈ 90 °C for 10 min in the oven for drying and stabilizing.

Material Characterization: The surface morphology of the sensing materials was characterized by a field emission scanning electron microscope (SEM) (Utral 55, ZEISS, Germany). The elemental analysis of the MOF was gained by an angular resolution X-ray photoelectron spectrometer (Kratos Analytical Ltd, UK). Fourier-Transformed Infrared (FT-IR) spectra were obtained by a Fourier infrared spectrometer (Bruker Scientific Instruments, Germany). The adsorption/desorption performance and specific surface area were tested by Brunauer–Emmett–Teller (BET) analysis, and the micropore size and distribution were tested by Barret–Joyner–Halenda (BJH) analysis with pore size analyzer (ASAP2020HD88, Micromeritics Instruments Corp., USA).

Sensing Measurement: Before the sensing measurement, the MOF-coated QCM device was connected to the developed frequency shift detector made of PCB. The output of the proposed system was collected by a microcontroller (Arduino Uno), and transmitted to an upper computer for reading, display, and storage by a python program. All experiments were carried out at room temperature (25 °C).

Statistical Analysis: The baseline of the voltage signals obtained from all experiments was adjusted to the same value, thereby facilitating a better comparison of the frequency response of different reagents under test. All experimental data were depicted as mean values $\pm$ standard deviation from three independent measurements. The serial data of the microcontroller was collected by a Python program with an interval of 1 s. MATLAB was used for data processing and calculation, and OriginLab was used for data fitting, statistical analysis, and plotting.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

Acknowledgements

This work was supported by the National Natural Science Foundation of China (under Grant U21A20519) and the Key Technology Research and Development Program of Zhejiang Province (under Grant 2021C03177).

Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

frequency shift detection, mass detection, phase-locked loop (PLL), quartz crystal microbalance (QCM)

Received: September 29, 2024

Revised: December 16, 2024

Published online:

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